Naga Venkata Sai Nadh Vaka

M.Eng. Student | RTL Design | Digital Verification | FPGA Implementation

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PROFILE SUMMARY

Master's student in Electrical and Microsystems Engineering at OTH Regensburg focused on digital IC design and semiconductor technologies. Hands-on experience in Verilog RTL design, functional verification, simulation, synthesis, and FPGA implementation, with programming experience in Python and academic foundations in CMOS/VLSI and circuit simulation. Interested in extending this foundation into SPICE simulation and simulation-flow automation.

EDUCATION

03/2026 – Present
Regensburg, Germany

Master of Engineering in Electrical and Microsystems Engineering,
Ostbayerische Technische Hochschule (OTH) Regensburg

12/2021 – 04/2025
Vijayawada, India

Bachelor of Technology in Electronics and Communication Engineering,
Usha Rama College of Engineering and Technology

- **Final Grade:** 1.6 (Graduated with Distinction)
- **Qualified GATE 2024** (EC)

TECHNICAL SKILLS

RTL & Digital Design: Verilog, VHDL, SystemVerilog , RTL Design, FSM Design, Combinational & Sequential Logic

Verification & FPGA: Verilog Testbenches, Functional Verification, Simulation, Synthesis, FPGA Implementation, Timing Analysis, AMD Vivado

Programming & Scripting: Python, C, MATLAB, Git, Keil µVision, Proteus, Arduino IDE

Interfaces & Hardware: AXI4-Lite, SPI, UART, I^2C, Microcontrollers, Sensor Interfaces, PCB Prototyping

Languages: English (C1), German (A2), Telugu (Native)

EXPERIENCE

11/2024 – 04/2025
Vijayawada, India

Embedded Systems with IoT Applications Intern, *Protective Technologies Pvt. Ltd.*

- Supported technical documentation and functional system testing while working with UART/I^2C/SPI peripherals.

08/2024 – 04/2025
Vijayawada, India

President - ECRATZ ECE Student Association, *Usha Rama College of Engineering and Technology,*

- Coordinated technical activities, project milestones and communication for 700+ engineering students under deadline-driven conditions.

PROJECTS

AXI4-Lite SPI Flash Controller IP Core, *Verilog | AMD Vivado*

- Designed a synthesizable AXI4-Lite slave and FSM-based SPI controller for memory-mapped SPI NOR Flash.
- Integrated register decoding, SPI sequencing, 24-bit addressing and Flash command execution in Verilog RTL.
- Developed an AXI4-Lite master testbench and SPI Flash behavioral model for functional verification and data readback.
- Implemented on Xilinx Virtex-7 using AMD Vivado: 193 LUTs, 206 FFs, 0 timing failures at 100 MHz, FMAX ≈ 212.8 MHz.

Digital Logic Design using Verilog & VHDL, *Bachelor's Laboratory Projects*

- Developed combinational and sequential digital circuits including multiplexers, encoders, flip-flops, counters, and shift registers.
- Developed custom testbenches and verified functionality through simulation and synthesis.

Transformerless AC-to-DC Power Supply, *Electronics & Laboratory Testing*

- Designed and tested a multi-output 5V/9V/12V/15V power-supply prototype; performed electrical measurements and troubleshooting.

PUBLICATIONS

04/2025

Advanced Image Encryption Algorithm Integrating Chaotic Image Encryption and CNN,
i-manager's Journal on Electronics Engineering